

Table of contents

Interdependencies of Fields: Electro-Magnetic System (Maxwell conditions) . . . 1

Interdependencies of Fields: Electro-Magnetic System (Maxwell conditions)

Static, kinetic rest charge localisation (conservative system):

...

Stationary, slow charge movement localisation (conservative system):

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \quad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t} \mathcal{B} = i\omega \mathcal{B} \quad \nabla \times \frac{1}{\mu_0} \mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t} \mathcal{E} = j - i\omega \varepsilon_0 \mathcal{E} \quad (1)$$

Dynamic, fast charge movement (relativistic):

...